

Playground xyz: Reinventing attention measurement with Google Cloud AI capabilities to improve ad success

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ABOUT PLAYGROUND XYZ



About Playground xyz

Founded in 2015, Playground xyz is a marketing technology company on a mission to master the art and science of maximizing consumer attention. The company's best-in-class technology stack, the Attention Intelligence Platform, powers a comprehensive suite of tools to help advertisers create, measure, and optimize truly attention-grabbing ads. Playground xyz has seven offices across the world.

Industries: Technology

Playground xyz is effectiveness using tracking panel data Vertex AI machine models on BigQuery Cloud Run to acc

predict consume
Time on ads at s

Location: Australia

Google Cloud results

- Analyzes millions of impressions per hour across the web
- Responds to more than half a million bid requests per second from ad networks in less than 10 milliseconds
- Delivers ad hoc reporting in seconds, for quick marketing decisions

Helps
Advertisers
with 9
accuracy

(<https://cloud.google.com/cloud-console/>),
Google Cloud (<https://cloud.google.com/>)

[Cloud Natural Language API](https://cloud.google.com/natural-language/)

App Engine (<https://cloud.google.com/appengine/>)

Cloud Run (<https://cloud.google.com/run/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

[Dataflow](https://cloud.google.com/dataflow/), [Cloud Storage](https://cloud.google.com/storage/)

Google Cloud console

(<https://cloud.google.com/cloud-console/>)

(<https://cloud.google.com/products/operations>)

Cloud CDN (<https://cloud.google.com/cdn/>)

Cloud Natural Language API

(<https://cloud.google.com/natural-language/>)

Let us take on your challenge. We're here to help.

Cloud Run (<https://cloud.google.com/run/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Operations

(<https://cloud.google.com/products/operations>)

To catch consumers' attention, advertisers often bid for ad placements in websites and apps based on “viewable” impressions. The idea behind this is that if an ad appears on the screen, it has a chance to make an impact on the viewer. However, just because an ad is viewable, does not necessarily mean someone looked at it. To solve this problem, ad tech company Playground xyz (<https://playground.xyz/>) pioneered its eye-tracking algorithm to predict Attention Time: a next generation metric which accurately determines how long, in seconds, someone actually looks at an ad.

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—Jonathan Mullins, CTO
& co-founder, Playground
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With the help of Playground xyz’s award-winning Attention Intelligence Platform (AIP), advertisers can track Attention Time—the amount of time people spend looking at an ad—to make bidding decisions more effectively. Playground xyz found that higher attention correlates not only with better brand building, but also higher conversions and sales. Powered by Google Cloud (<https://cloud.google.com/>), AIP comprises a comprehensive suite of tools which use machine learning (ML) intelligence to create, measure, and optimize ads that secure the Attention Time an ad receives.

“We’re leaning on Google Cloud to deliver attention measurement and bid optimization at scale for advertisers around the world,” says Jonathan Mullins, co-founder at Playground xyz. “We analyze

millions of impressions per hour across the web and respond to more than half a million bid requests per second from ad networks in less than 10 milliseconds.”

Playground xyz hosts its application on [App Engine](#) (/appengine) and uses [BigQuery](#) (/bigquery), [Vertex AI](#) (/vertex-ai), [Dataflow](#) (/dataflow), and [Pub/Sub](#) (/pubsub) to drive its attention-measurement technology.

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Accurately predicting consumer attention with real-world gaze data

Getting started with ML can be challenging for ad tech companies with lean engineering resources. To overcome this, Playground xyz uses Vertex AI to simplify ML workflows. The company uses eye gaze data from more than 10,000 opt-in eye-tracking research participants to train its ML prediction model for Attention Time. The dataset consists of X,Y coordinates of the participant’s eye gazes as they read a variety of articles and content types, without saving images or personal data. Along with the eye gaze data, the company tracks 40 other data signals such as content type, page geometry, scrolling speed, and ad format characteristics.

The AI matches the eye-tracking data with the 40 other data signals to create a model of predicted Attention Time. This allows the AI to measure

Attention Time out on the open web. “Our Vertex AI model predicts the average Attention Time with 94% accuracy compared to holdout sets, so customers can be confident about this extrapolated score,” says Mullins. “Dataflow and BigQuery deliver reliable data pipelines, making it incredibly easy for our engineers to run and deploy new models that scale.”

Analyzing campaign data for digital advertising insights in seconds

Playground xyz knows the success of its AIP lies in the accuracy and speed of its analytics dashboard. Advertisers rely on real-time analytics to understand how consumers are interacting with their ads across their media buying. To help advertisers make informed decisions about what ad to place and where, Playground xyz relies on BigQuery as its analytics platform to keep up with data volume as ads and media channels increase.

To deliver data-driven insights, BigQuery gathers ad performance data across publisher websites each day and uses Vertex AI to measure each impression. AIP then scores the Attention Time for the ad on the page and calculates the optimal bid based on the result.

By doing this, Playground xyz provides the feedback loop to help advertisers identify where a specific creative is getting more attention. Based on the data insights, advertisers can set up automated workflows in AIP to target those high-performing

channels, such as a workflow to only bid on articles that generate the top 30% of Attention Time.

BigQuery analyzes ten petabytes of raw data per month, directly from AIP stored in Cloud Storage (/storage). Instead of paying a fixed monthly subscription, Playground xyz optimizes cost with on-demand billing for high volumes of events and queries. The company expects the data to scale exponentially, with the number of customers on the platform growing every week.



Delivering ad-tech workloads at speed with Cloud Run

Playground xyz's rich media ad server handles hundreds of thousands of requests per second from clients around the world, serving ad content to devices and tracking delivery and performance metrics. Any ad server downtime impacts advertisers and publishers alike. Advertisers won't be able to reach customers because their ads don't

appear on targeted websites. Similarly, publishers may lose ad revenue and customer confidence if ads are slow to load.

This is why the ad server is one of 12 business-critical applications that Playground xyz is migrating to [Cloud Run \(/run\)](#). Previously, Playground xyz ran the application on a legacy cloud provider, independently deploying to multiple regions to serve its global client base. The team had to switch between different regions to view operational metrics. Cloud Run provides built-in integration with Google Cloud's [operations suite \(/products/operations\)](#) so the team can monitor the health of the application with metrics such as application performance and response times. "The operations suite is a game changer. By creating alerts for critical application metrics and pushing notifications to the appropriate team, we're able to identify potential issues before they cause any service disruption," says Mullins.

Playground xyz uses [Cloud CDN \(/cdn\)](#) content delivery network with Cloud Run to help advertisers provide a seamless user experience for customers in different regions. To reduce the ad load time, Cloud CDN caches creative content in globally distributed servers. Once a consumer opens on a publisher's page, cached ad content loads from a Cloud CDN server closest to the end user instead of connecting to the ad server, to speed up ad delivery.

Mullins adds, "Google Cloud helps us allocate more engineering resources to value creation and new product development. It gives us confidence in the

scalability and availability of our applications as we grow, allowing us to focus on architecture instead of infrastructure.

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Transitioning from prototype to production

Google Cloud reduces the complexity of having multiple teams delivering high availability, high-traffic global applications. Mullins says his teams can easily set up self-contained projects on Cloud Run to test new prototypes and easily transition them into production when required.

The Playground xyz team is also working on an AutoML Natural Language model to recognize news headings on the publishers' websites. The intent is to match ads to the right context at the article level. By delivering more relevant ads, the company hopes to fight banner blindness and improve the consumer experience.

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"Ultimately, Google Cloud empowers us to better serve our customers, helping them create, measure, and optimize attention-grabbing ads which drive real business results."